

Quantifying Framing and Ideological Bias through Fine-Tuning in Large Language Models

Nils Neuner and Hamed Laroui

University of Konstanz

Abstract

This paper examines whether fine-tuning large language models (LLMs) on news coverage from media outlets can serve as a functional proxy for narrative adoption through media exposure. Combining media framing theory, survey methodology, and computational language modeling, we fine-tune Gemma 3-12B models on 980 articles from each of four Israeli and Palestinian outlets and measure changes in responses to survey-style propositions. A cross-averaged agreement score, based on responses to propositions and their negations, is used to reduce response artefacts. Across five dimensions of the Israeli-Palestinian conflict, the results show shifts that often correspond to outlet orientation, particularly regarding agency, military legitimacy, and responsibility for casualties. Differences between the two Israeli outlets suggest sensitivity to variation within an ideological camp, while several items produce convergence across outlets. We therefore find preliminary support for fine-tuning as a functional cognitive proxy, while emphasizing that it does not establish equivalence with human cognition. The findings suggest that LLM-based exposure experiments can help identify candidate items for measurement of media narratives, which should be validated with human respondents.

1 Introduction

Some of the most productive moments in academic research happen at the seams between disciplines, where a method built for one purpose ends up answering a question from somewhere else entirely. This paper is an attempt at exactly that kind of borrowing, bringing together three strands of literature, media framing theory, survey-based attitude measurement, and computational language modeling, to find a new way to capture media influence on human narrative adoption.

Framing and narrative theory in media studies offers a rich account of how news coverage shapes the way audiences interpret events, but lacks an easy way to measure that influence quantitatively: narrative effects tend to be subtle and cumulative rather than immediate, so they resist direct empirical measurement. Survey methodology and psychological measurement, by contrast, offer exactly the toolkit framing research is missing. Decades of work have produced reliable instruments for measuring attitudes and narrative alignment in human subjects. But this toolkit has an obvious constraint: it depends on having a subject who can respond. A survey can be administered to a person; it cannot be administered to a news article.

We propose closing this gap with an approach borrowed from computational social modeling. If an LLM is fine-tuned on the output of a specific news outlet, it can be treated as a cognitive proxy for a reader habitually exposed to that outlet’s coverage, a stand-in that,

unlike the article itself, can actually be asked a question. The logic is straightforward: if, after fine-tuning, the model answers a survey item differently, or comes to endorse a narrative it previously did not, that shift can reasonably be attributed to narrative content in the outlet’s news corpus.

This leads to the first research question:

RQ1: To what extent can fine-tuning a large language model on the media output of a particular news outlet function as a cognitive proxy for narrative adoption through media consumption?

To test this proposition, we apply it to a case study defined by competing narratives: the Israeli-Palestinian conflict. We select a set of Palestinian and Israeli media outlets and fine-tune a separate model on each, comparing how exposure to each outlet’s coverage shifts the model’s responses relative to a common baseline.

Within this broader conflict, we focus on the victim-aggressor cleavage: the competing ways media narratives assign responsibility, legitimacy, and victimhood to the actors involved. This cleavage can manifest through different framing dimensions, including who is portrayed as responsible for harm, whether military actions are considered legitimate, and how each side’s suffering is represented.

This case study lets us investigate whether shifts in fine-tuned models reveal which aspects of this cleavage

are most strongly reflected in different media environments. We therefore ask:

RQ2: What can the shifts in fine-tuned models reveal about the structure of the victim-aggressor cleavage in coverage of the Israeli-Palestinian conflict, and which narrative dimensions provide the strongest basis for its measurement?

A project of this scope draws on several distinct bodies of literature, and the paper is structured to walk through each in turn. Section 2.1 lays out the conceptual foundations of framing and narrative in media studies. Section 2.2 develops the idea of using LLM fine-tuning as a cognitive proxy for narrative adoption. Section 2.3 turns to the specific case, surveying the framings and narratives existing literature has identified within Israeli-Palestinian conflict coverage. Section 3 provides the paper’s methodological core: an overview of the pipeline connecting fine-tuned models to survey-style questioning, along with a discussion of how to address practical issues such as filtering out noise introduced by the fine-tuning process itself.

2 Literature and Concepts

2.1 Narratives and Framing

This paper follows the approach of Gamson and Modigliani [1], which conceptualized framing as a tool to construct narratives that align with the broader ideological and institutional setting of the media outlet. Media companies are thus not considered as objective actors but influenced by economic, political, and institutional forces that shape the ideological footprint of the media [2], which then manifests itself in the employment of certain narratives. Such narratives are defined as overarching stories or patterns of meanings [3], which can be constructed via framing by selecting or emphasizing certain aspects of a perceived reality, making them more salient while marginalizing others [4]. For the consumer of media, these frames can serve as schemata of interpretation which enables the consumer to perceive, identify and categorize events and information [5] and can indirectly lead to the adoption of the constructed narratives by the reader.

The present contribution draws attention to the subtle nature of frames: individually, they may go unnoticed, yet their cumulative effect over time enables an indirect transfer of narrative to the audience, one that operates without the underlying narratives being explicitly articulated or consciously perceived by the reader. This subtlety renders quantitative empirical analysis particularly demanding. Beyond the already considerable difficulty of developing objective measurement techniques

for framing, it remains an open question how the cumulative impact of such frames on the construction of broader narratives can be reliably identified and assessed. To address this challenge, we adopt an approach in which the adoption of narratives by a media consumer is simulated through fine-tuning a large language model on the articles of a given media outlet. An in-depth discussion of this provided in Section 2.2.

2.2 Fine-Tuning as a Cognitive Proxy for Narrative Adoption through Media Consumption

A full analytical comparison between human media consumption and LLM fine-tuning would require examining both the processes through which exposure produces observable changes and the mechanisms through which those changes are learned. That comparison is beyond this paper’s scope. We instead focus on the observable relationship between exposure and subsequent responses, asking not whether fine-tuning reproduces the psychological mechanisms of human media consumption, but whether exposure to ideologically distinct corpora produces model output shifts analogous to those documented in human media-exposure research: a functional rather than mechanistic equivalence.

Prior work establishes this progressively. Early studies show fine-tuning on politically biased data shifts LLM ideological outputs [6][7], but rely on instruction-tuned tasks (e.g., stance ranking), leaving open whether the shift reflects genuine narrative adoption or task-specific learning. More direct evidence comes from studies that fine-tune blindly on raw corpora and validate against human data:

Media-diet models, fine-tuned on the news, television, or radio content consumed by particular populations, reproduce patterns in nationally representative survey responses, with predictive performance increasing among populations that pay greater attention to the corresponding media [8]. The models thus come to reflect narratives embedded in the media environment itself, an empirical link between model adaptation and human media effects that goes beyond merely reproducing surface features of the source texts. This approach has since extended from general media diets to explicitly political communities: fine-tuning Llama Chat on the speeches of individual European political parties shifts the models’ positions towards those parties when evaluated on the EUandI political-attitude questionnaire [9]. The approach thus recovers narrative framing specific to a political community from naturally occurring discourse alone, without requiring explicit instructions about which frame to adopt [10].

The strongest evidence for a distinction between such community-level adaptation and simple memorization comes from recent work on epistemic stance transfer.

Gerard, Chang, and Volkova [11] fine-tune models on discourse from distinct online communities, then deliberately delete the models’ knowledge of the events represented in the training material. Despite this targeted deletion, the models continue to exhibit community-specific patterns in how they respond to new uncertainty, across both Russian-Ukrainian military discourse and U.S. partisan Twitter data. This matters because the models’ behavior persists precisely when reproducing previously learned facts can no longer explain it: the models appear to retain a generalized interpretive schema for responding to information and uncertainty tied to a given community, not just a memorised record of its claims.

Together, these three findings trace a progression from content adaptation to narrative transfer, and provide the strongest basis for treating fine-tuning as a functional analogue of sustained exposure: the relevant outcome is not that the model remembers the corpus, but that exposure reshapes the frames through which it interprets information outside it.

2.3 Narratives and Framing applied to the Israel-Palestine Conflict

Having established both the theoretical logic and empirical precedent for using LLM fine-tuning as a proxy for narrative adoption, this section applies these tools to the Israel-Palestine conflict. The conflict suits the inquiry well: it is structured by a deeply entrenched ideological cleavage that produces competing, incompatible narrative framings, making it an ideal testing ground for whether fine-tuned models absorb and reflect these divergent perspectives. Precisely because these framings are so entrenched and mutually incompatible, however, the conflict resists analytical summary without some oversimplification, and the framework offered here does not claim to capture its full complexity or represent all competing narratives with equal depth.

Broadly, Palestinian actors and political positions uphold the principle of self-determination and frame their struggle as resistance against a colonial, occupying power, while Israeli and Zionist discourse foregrounds the historical, political, and religious claim to statehood, the state’s sovereign legitimacy, and the necessity of security measures against perceived existential threats. Such divergent framings can produce representations of Palestinians primarily as a security concern or existential threat to the Israeli state [12].

This raises the question of how these competing narratives are reflected in media coverage. Rather than examining international coverage, this study focuses on outlets situated within the Israeli and Palestinian media environments themselves, allowing us to examine how competing narratives are constructed and expressed by media directly associated with each side. Future re-

search could examine how these narratives are subsequently reflected in international coverage. We therefore turn to the specific narrative structures and framing techniques identified in the literature.

The literature identifies two such narratives, both employable by either side. The “victim versus aggressor” narrative is constructed through several techniques, each pointing to a distinct dimension along which coverage can be characterized. First, passive versus active sentence structures can obscure or assign responsibility: “Two Palestinians died in an airstrike” removes the agent, whereas “An Israeli airstrike killed two Palestinians” explicitly attributes agency [13]. At stake here is both the attribution of deliberate intent and the attribution of responsibility and blame for casualties, since the same grammatical choice determines whether harm appears as something an actor did or as something that merely happened. Second, word choice can shift perceptions of responsibility, for example by describing an airstrike as a “preventive measure” or “defensive military operation” [14], framing that bears directly on the justification and legitimacy of military action. Third, unequal depictions of suffering, such as giving one side personal testimonies or graphic imagery while presenting the other through statistics, can create a hierarchy of victimhood by making one side’s losses more salient. Beyond these explicit choices, selective omission, unequal coverage, and reliance on one party’s official sources may further reinforce the victim-versus-aggressor narrative by systematically privileging certain actors, events, or interpretations.

A related “us versus them” narrative is particularly relevant for international media coverage. Here, audiences are positioned as external observers who may be encouraged to identify with one side as an in-group while distancing themselves from the other as an out-group [15]. For example, pro-Israeli discourse may describe Israel as “the only democracy in the Middle East” and Hamas as “terrorists,” while pro-Palestinian discourse may use “Free Palestine,” “Zionist state,” or “Israeli Occupation Forces” (IOF) to emphasize Palestinian liberation and Israeli occupation. Such terminology can construct competing group identities and confer or withhold political legitimacy. This framing is less directly applicable here, however, since the study examines outlets originating from the two sides of the conflict rather than international outlets addressing an external audience. Our analysis therefore focuses exclusively on the direct Israeli-Palestinian cleavage, operationalized through the victim-versus-aggressor narrative.

Taken together, these techniques suggest five dimensions along which outlet coverage can be characterised: agency attribution, legitimacy of military action, responsibility for casualties, political terminology, and comparative victimhood, taken up in Section 3.6 as the basis for our measurement instrument.

3 Methods

Having established the theoretical rationale for treating fine-tuning as a proxy for narrative adoption, and having situated the Israel-Palestine conflict as a case in which this adoption can be meaningfully studied, we now turn to how this framework is implemented in practice. The remainder of this section is organized in three parts. Section 3.1 describes the data collection process, detailing which outlets were selected and the criteria that governed this selection. Section 3.2 lays out the data-processing pipeline, and Section 3.3 covers the fine-tuning procedure while Sections 3.4 and 3.5 explain the methods used to measure resulting shifts in model agreement. Section 3.6 then addresses the operationalization strategy: how the narratives identified in the literature are translated into a concrete set of measurable items.

3.1 Data Collection

Outlet selection was governed primarily by practical rather than ideological criteria: an outlet had to publish in English, since the fine-tuning and probing procedure operates entirely in English, and its website had to permit scraping at the required volume.

Our initial aim was broader: to scrape a wider set of outlets from both countries representing their respective media landscapes more fully, ideally with at least one public broadcaster, one outlet positioned toward the extreme of the conflict cleavage, and one more moderate outlet per country. The two criteria above, however, narrowed this considerably, leaving two outlets per country:¹

- **Israel National News (INN)** is a privately owned Israeli outlet operated under the Beit El yeshiva center, associated with the religious Zionist and settler movement.
- **The Jerusalem Post** is a privately owned, commercially operated Israeli newspaper with a broader, more mainstream readership.
- **Wakālat al-Anbā’ al-Filasṭīniyya, the Palestinian News Agency (WAFA)** is the official news agency of the Palestinian Authority/PLO, operating under the PLO Executive Committee.
- **Palestine News Network (PNN)**, based in Bethlehem, is an independent Palestinian outlet run by a collective of journalists, publishing in English, Arabic, and Hebrew.

From each outlet, articles are collected from 7 October 2023, the onset of the current escalation, onwards,

¹Scripts and code: Quantifying-Framing-and-Ideological-Bias-through-Fine-Tuning-in-LLMs

and retained if they contain at least one term from a predefined conflict-relevant keyword list (Appendix A).

This composition falls short of the balanced distribution originally envisioned. WAFA provides a public, state-affiliated outlet on the Palestinian side, but no comparable English-language public broadcaster could be identified for Israel. On the Israeli side, the two outlets span a meaningful range along the conflict cleavage, with The Jerusalem Post representing a more moderate position and Israel National News a more extreme one; the two Palestinian outlets are considerably harder to differentiate in stance. This asymmetry is compounded by uneven coverage in existing bias-rating infrastructure: Media Bias/Fact Check (MBFC) [16] rates Israel National News and the Jerusalem Post as Right-Center, while AllSides [17] rates the Jerusalem Post as Center. MBFC rates PNN as Left, with a Mixed factual-reporting assessment, but WAFA is not covered by either MBFC or AllSides, leaving one of the four outlets without an external benchmark against which its editorial position can be independently verified. This uneven balance is an important limitation addressed at the end of the paper.

3.2 Data Processing Pipeline

As outlined in the introduction, our approach treats the fine-tuned model as a cognitive proxy capable of answering survey-style questions, allowing us to measure its degree of agreement with a given proposition in the same way a survey would measure a respondent’s attitude. Since our interest lies specifically in how this agreement shifts as a result of fine-tuning, we develop a pipeline that estimates, for each item, the change in agreement induced by fine-tuning, computed separately for each outlet-specific model. Figure 1 provides an overview of the procedure. The remainder sections describe its three stages: (1) obtaining the baseline and outlet-specific fine-tuned models, (2) eliciting item-level agreement scores and correcting for measurement artefacts introduced by the probing procedure, and (3) computing the change in agreement between the baseline and fine-tuned models.

3.3 Fine-Tuning Procedure (1)

For each outlet, we fine-tune an independent copy of the same open-weight base model, Gemma 3-12B [18], which contains approximately 12 billion parameters. Fine-tuning is performed using Low-Rank Adaptation (LoRA) [19], a parameter-efficient fine-tuning method that introduces trainable low-rank weight updates while leaving the original model parameters frozen. To reduce the computational and memory requirements of training, LoRA is combined with 4-bit quantization (QLoRA) [20].

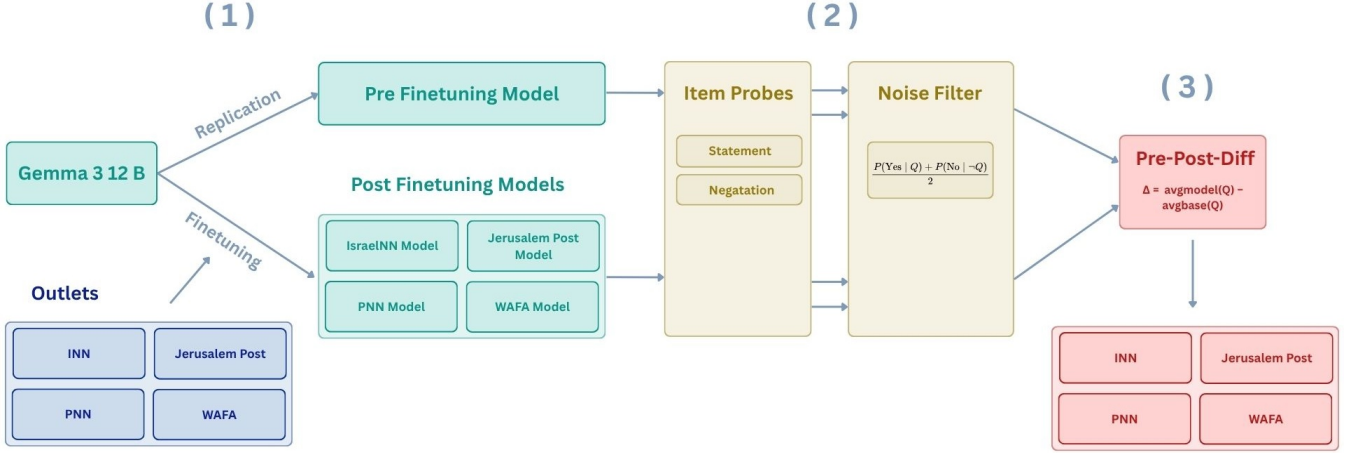


Figure 1: Overview of the data-processing pipeline. (1) The base model, Gemma 3-12B, is independently fine-tuned on each outlet’s corpus to produce four outlet-specific models. (2) Each model is probed on the item bank (statement and negation pairs), and (3) the paired responses are used to compute a noise-corrected, cross-averaged agreement score, from which the pre-to-post-fine-tuning shift $\Delta(Q)$ is derived for each item on each outlet.

All outlet-specific models are initialized from the identical base model and trained using the same configuration. The complete set of hyperparameters is reported in Appendix C. Consequently, the composition of the training corpus constitutes the principal experimental difference between fine-tuning runs. This design allows subsequent differences in model responses to be interpreted relative to the respective outlet corpus rather than to differences in model initialization or training procedure.

Because the outlets differ substantially in publication volume, their corpora are first downsampled to a common size of 980 articles, corresponding to the smallest outlet corpus and we normalize article length. Each article is represented by its title and full body text and is provided to the model as plain training text. Five percent of the articles from each outlet are held out from the training set and used to monitor training progress.

3.4 Measurement Instrument (2)

For each item, we use a fixed prompt that requires the model to express its response using a discrete numerical token. Rather than sampling generated responses, we extract the model’s next-token probability distribution over the relevant response tokens and normalize these probabilities using the softmax function. This procedure yields a continuous measure of the model’s response tendency which can be interpreted as the probability that the model agrees with the item.

Response noise and correction However the raw probing probabilities can be misleading for two reasons. First, the base model may have no clear stance on a

proposition and nevertheless tend to answer “No.” In that case, it may assign high probability to “No” both for a proposition Q and for its negation $\neg Q$. This should not be interpreted as disagreement with Q : a model that genuinely disagrees with Q should instead answer “No” to Q and “Yes” to $\neg Q$. Answering “No” to both indicates that the model is not committing to either position.

Second, fine-tuning can change the model’s probability distribution without changing its substantive stance. Probability mass may become more dispersed across response options, causing a response that previously had probability close to zero to receive a small positive probability after fine-tuning. If we measured agreement using only $P(\text{Yes} | Q)$, this redistribution could therefore appear as a shift toward agreement even when no such shift has occurred.

The problem is illustrated by Q4 in Figure 2. Before fine-tuning, the model assigns essentially zero probability to “Yes” for both $Q4$ and $\neg Q4$. After fine-tuning, these probabilities increase to 0.042 and 0.203, respectively. Looking only at $P(\text{Yes} | Q4)$ would suggest that the model has become more agreeable to $Q4$. However, this conclusion is unwarranted: the base model’s responses to $Q4$ and $\neg Q4$ indicate non-commitment rather than disagreement, while the small post-fine-tuning probability can arise simply from the broader response distribution. We therefore evaluate each proposition together with its negation rather than interpreting either response in isolation.

Cross-averaged agreement score To reduce the influence of these response artefacts, we construct a symmetric agreement measure using both the proposition Q

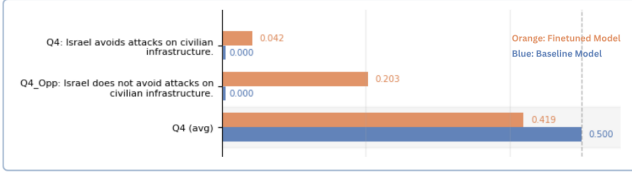


Figure 2: Example of fine-tuning noise (item Q4): raw $P(\text{Yes})$ alone suggests a shift toward agreement, while the paired statement/negation measure reveals the opposite.

and its logical negation $\neg Q$. Specifically, we define the cross-averaged agreement score as

$$\text{avg}(Q) = \frac{P(\text{Yes} \mid Q) + P(\text{No} \mid \neg Q)}{2}.$$

The two terms capture complementary aspects of agreement. $P(\text{Yes} \mid Q)$ measures the probability that the model affirms the proposition in its original formulation, whereas $P(\text{No} \mid \neg Q)$ measures the probability that the model rejects the logically equivalent negative formulation. A high value therefore requires the model to respond consistently with agreement across both formulations.

This produces an intuitive scale from 0 to 1. A value of 1 indicates consistent agreement with Q , while 0 indicates consistent disagreement. Importantly, a “No” response from the model is not necessarily a problem. When the model answers “No” to both Q and $\neg Q$, the two terms in the equation are close to 0 and 1, respectively, giving

$$\text{avg}(Q) \approx 0.5.$$

We interpret this midpoint as neutrality or non-commitment rather than disagreement. In fact, the model’s frequent “No” responses are useful under this measure: they provide a natural neutral point exactly halfway between agreement (1) and disagreement (0). This makes changes in model stance particularly straightforward to interpret. For example, a shift from 0.5 to 0.7 represents a movement from neutrality toward agreement, while a shift from 0.5 to 0.3 represents a movement toward disagreement.

By contrast, genuine disagreement with Q requires the model to respond “No” to Q and “Yes” to $\neg Q$. Both terms in the equation are then close to zero, yielding $\text{avg}(Q) \approx 0$. Genuine agreement produces the opposite pattern, with both terms close to one and $\text{avg}(Q) \approx 1$.

The cross-averaged score therefore provides a simple and symmetric measure of agreement that distinguishes substantive positions from a generic “No” response. We use this score as the item’s agreement measure throughout the analysis.

For Q4, the base model yields

$$\text{avg}_{\text{base}}(Q4) = 0.500,$$

indicating neutrality. After fine-tuning, the score decreases to

$$\text{avg}_{\text{model}}(Q4) = 0.419.$$

Thus, rather than the apparent shift toward agreement suggested by the raw probability of “Yes”, the paired measure indicates a modest shift from neutrality toward disagreement.

3.5 Delta / Shift (3)

In the last step, we quantify for each model the change in agreement relative to the common base model on an item-by-item basis. The primary outcome measure is defined as

$$\Delta(Q) = \text{avg}_{\text{model}}(Q) - \text{avg}_{\text{base}}(Q).$$

Accordingly, $\Delta(Q) > 0$ indicates that fine-tuning increased the model’s agreement with proposition Q , whereas $\Delta(Q) < 0$ indicates a shift toward disagreement. Values close to zero indicate little or no change in the model’s measured stance.

The resulting $\Delta(Q)$ values constitute the primary unit of analysis for comparing the effects of outlet-specific fine-tuning. Because all outlet-specific models originate from the same base model and the agreement measure is computed using the same paired-proposition procedure, the resulting shifts can be interpreted as changes associated with the respective outlet corpora rather than as differences in the models’ initial response behavior.

3.6 Construction of the Item Bank

As one of the first attempts to use LLM fine-tuning as a proxy for narrative adoption, the present study is necessarily exploratory. We therefore do not aim to establish a definitive measurement instrument for the Victim-versus-Aggressor cleavage at this stage. Instead, we use the analysis as a preliminary investigation of how this cleavage manifests in model responses, identifying promising dimensions and items that can provide a foundation for future research. Subsequent studies can build on these findings to develop, refine, and ultimately validate a construct for measuring the Victim-versus-Aggressor cleavage.

The theoretical framework identifies the Victim-versus-Aggressor narrative as a central structure through which the ideological cleavage surrounding the conflict may manifest in media coverage. Based on the literature and conceptual discussion in Section 2.3, we identify five dimensions through which this narrative may be expressed: (A) attribution of deliberate intent and agency, (B) justification and legitimacy of military action, (C) attribution of responsibility and blame for

casualties, (D) terminology and the legitimacy of political labels, and (E) comparative suffering and the construction of victimhood hierarchies. Each dimension is represented by directional propositions designed to capture a different aspect of how actors may be positioned as victims or aggressors. The complete item bank is provided in Appendix D.

These five dimensions should be understood as theoretically motivated candidate dimensions rather than established components of a single construct. At this stage, it is not known whether all five dimensions capture the same underlying cleavage, whether some represent distinct aspects of conflict framing, or whether some are more suitable for measuring the Victim-versus-Aggressor cleavage. The purpose of the exploratory analysis is therefore to investigate how individual propositions respond to exposure to contrasting media environments and identify patterns that can inform subsequent construct development.

This approach follows the general logic of survey scale development, in which a broader construct is represented through multiple items intended to capture related manifestations of an underlying attitude. Before such items can be combined into a scale, however, it is necessary to establish which items meaningfully distinguish the construct and whether they behave coherently. In the present study, the fine-tuned models provide a preliminary means of exploring this question. Rather than assuming that all theoretically derived items contribute equally to the Victim-versus-Aggressor cleavage, we examine the direction and magnitude of shifts produced by outlet-specific fine-tuning.

4 Results

We report results dimension by dimension, comparing the four outlet-specific models against the shared base model. Recall from Section 3.5 that $\Delta(Q) > 0$ denotes a shift toward agreement with proposition Q following fine-tuning, and $\Delta(Q) < 0$ a shift toward disagreement. Throughout, Wafa and PNN denote the Palestinian-outlet models while INN (or IsraelNN) and The Jerusalem Post shape the Israeli-outlet models.

For many of the items the observed shifts follow the expected direction: outlet-aligned models move toward their side’s framing and the two groups diverge. However, for some elements, the picture is more textured, and recurring patterns open up new interpretive questions. We first present the dimension-level results, then turn to these cross-cutting observations.

4.1 Deliberate Intent and Agency Attribution

Fig. 3 reports results for Dimension A. The expected pattern is clearly present. Both Palestinian-outlet mod-

els shifted toward agreement that Israel deliberately targets healthcare infrastructure (A1: Wafa +0.080, PNN +0.072), intentionally strikes journalists (A3: +0.080, +0.066), and intentionally restricts humanitarian aid (A5: +0.145, +0.139), while the Israeli-outlet models remained essentially unmoved on A1 and A3 (all $|\Delta| \leq 0.017$). Conversely, the Israeli-outlet models shifted strongly toward the human-shields proposition (A2: IsraelNN +0.361, Jerusalem Post +0.234) and toward deliberate targeting of Israeli civilians by Palestinian militants (A4: +0.437, +0.382), where the Palestinian-outlet models moved only marginally.

Two results depart from this pattern. First, the Palestinian-outlet models also shifted toward agreement on A4 (+0.083, +0.095), covering the claim the Palestinian militants deliberately target Israeli civilian in their attacks. Second, both Israeli-outlet models shifted toward agreement that Israel restricts humanitarian aid (A5: +0.046, +0.063), a proposition aligned with the opposing narrative. We return to both in Section 4.6.

4.2 Justification and Legitimacy of Military Action

Dimension B (Fig. 4) produces the sharpest outlet-level divergence in the study. The Israeli-outlet models shifted substantially toward accepting Israel’s campaign as legitimate self-defense (B1: IsraelNN +0.324, Jerusalem Post +0.279), while the Palestinian-outlet models barely moved (Wafa -0.007, PNN +0.022). The mirror image appears on disproportionality: the Palestinian-outlet models shifted strongly toward agreement that Israel’s actions constitute disproportionate and unjustified force (B2: +0.285, +0.308), where the Israeli-outlet models split, with IsraelNN moving modestly toward agreement (+0.088) and the Jerusalem Post slightly against (-0.035).

The blockade item separates the two Israeli outlets most clearly. IsraelNN shifted markedly toward endorsing the blockade as a necessary security measure (B5: +0.252), while the Jerusalem Post moved only marginally (+0.025); both Palestinian-outlet models shifted away (-0.053, -0.083).

Surprisingly, the October 7 items yield no interpretable signal. All four models shifted away from the proposition that the attack was a legitimate act of resistance (B3: -0.049 to -0.083), and all four also shifted away from the proposition that it was an act of terrorism with no legitimate justification (B4: -0.041 to -0.115). Since B3 and B4 stake out opposing positions, movement away from both cannot be read as a coherent stance shift in either direction, and we treat this item pair as inconclusive.

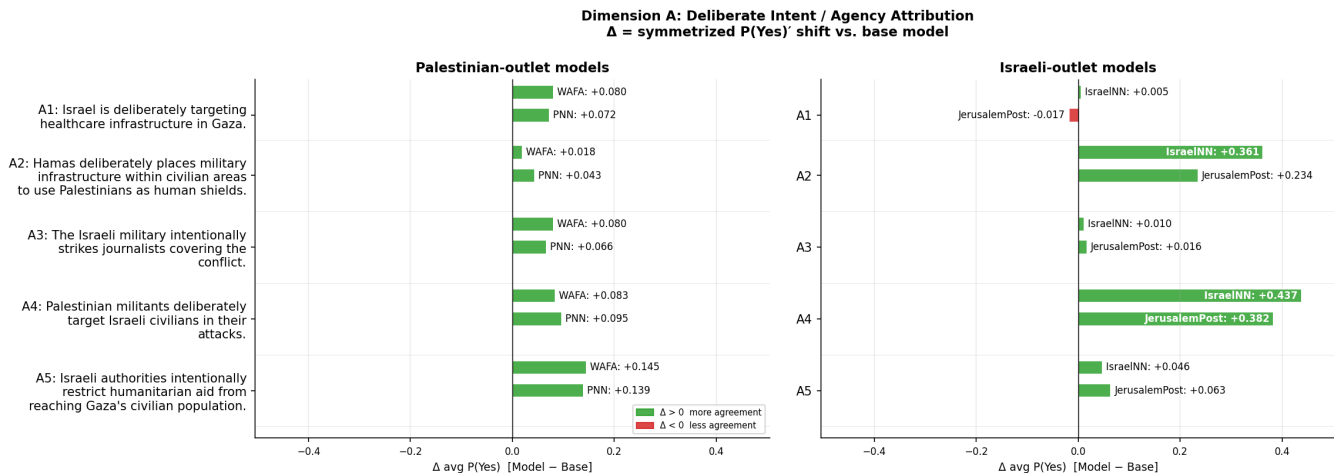


Figure 3: Dimension A (deliberate intent / agency attribution). $\Delta \text{ avg } P(\text{Yes})$: symmetrized shift in agreement, fine-tuned model minus base model, by item. Left: Palestinian-outlet models (WAFA, PNN). Right: Israeli-outlet models (Israel National News, Jerusalem Post). Green = increased agreement, red = decreased. A1, A3, A5 attribute harm to Israeli actors; A2, A4 to Palestinian/Hamas actors. Agreement increases for 19 of 20 model-item pairs; shifts are largest for Israeli-outlet models on A2/A4 (IsraelNN: +0.361, +0.437; JerusalemPost: +0.234, +0.382), well above their near-zero shifts elsewhere, while Palestinian-outlet models shift more evenly across all five items (+0.02 to +0.15).

4.3 Responsibility and Blame for Casualties

Dimension C (Fig. 5) separates the outlets along expected lines, but with a notable asymmetry in magnitude. Attribution of civilian deaths to Hamas (C1) moved sharply for IsraelNN (+0.416) and more moderately for the Jerusalem Post (+0.137), while both Palestinian-outlet models shifted marginally against it (-0.013, -0.029). A comparable split appears on Hamas’s use of civilian infrastructure (C4: IsraelNN +0.364, Jerusalem Post +0.198; WAFA +0.018, PNN +0.066). On the Palestinian side, the strongest movement concerns Israel’s failure to distinguish combatants from civilians (C3: WAFA +0.248, PNN +0.211) and primary responsibility for the humanitarian crisis (C5: +0.402, +0.346).

An important nuance is delivered by items C2 and C5: both Israeli-outlet models shifted toward agreeing on them (C5: IsraelNN +0.205, Jerusalem Post +0.152; and C2: +0.130, +0.033), attributing civilian deaths to the Israeli military’s conduct and claiming that Israel bears the primary responsibility for the humanitarian crisis in Gaza (+0.130, +0.033). This is especially surprising because the shift is even stronger for IsraelNN which is known for a more extreme stance on the conflict than the more moderate Jerusalem Post.

4.4 Labeling and Legitimacy Terminology

Dimension D (Fig. 6) yields clear confirmation on some items and surprising results on others. On D3, the

Palestinian-outlet models shifted strongly toward characterising Israel’s presence in the West Bank as an illegal occupation (WAFA +0.450, PNN +0.394), as expected. The Israeli-outlet response, however, is less straightforward: rather than shifting away, both models moved marginally in the same direction (+0.029, +0.052), and neither showed meaningful movement toward the competing lawful-presence framing (D4: IsraelNN +0.015, Jerusalem Post +0.014).

Furthermore, all four models, Palestinian and Israeli alike, shifted strongly toward agreement that Hamas is a terrorist organization (WAFA +0.379, PNN +0.368, IsraelNN +0.405, Jerusalem Post +0.372), with the Palestinian-outlet shifts nearly indistinguishable in magnitude from the Israeli ones. The corresponding resistance-movement item D2 shows essentially no movement in either direction for any model ($|\Delta| \leq 0.090$).

Most suprisingly, the genocide item (D5) produced negligible movement in either direction across all four models, leaving no clear stance on a proposition central to the contemporary framing contest.

4.5 Comparative Suffering and Victimhood Hierarchy

Dimension E (Fig. 7) splits into two parts. On proportionality, the pattern divides along expected lines: the Palestinian-outlet models moved toward the destruction in Gaza being disproportionate to the security threat (E3: +0.240, +0.235) and away from Israel’s response being proportionate (E4: -0.085, -0.100), while IsraelNN moved in the opposite direction on E4 (+0.162).

On the remaining three items, all four models converged regardless of outlet alignment. All shifted toward agreement that Palestinian civilian suffering has been greater than Israeli suffering (E1) and that Israeli suffering has been understated in media coverage (E2), with shifts ranging from +0.127 to +0.327 across both outlet groups. A similar, weaker convergence appears on E5, where all models shifted toward agreement that international coverage has failed to adequately represent Israeli victims. The victimhood dimension might thus produce cross-cleavage agreement on both the scale of Palestinian suffering and the inadequate representation of Israeli victims.

4.6 Patterns Across Dimensions

On the propositions where the ideological cleavage is sharpest, many observed shifts follow the expected direction. Palestinian-outlet models shifted toward agreement that Israel deliberately restricts humanitarian aid, that the military campaign constitutes a disproportionate use of force, that Israel bears primary responsibility for the humanitarian crisis, and that its presence in the West Bank constitutes an illegal occupation. Israeli-outlet models shifted toward endorsing the campaign as legitimate self-defense, attributing civilian casualties to Hamas’s use of human shields, and accepting the blockade as a necessary security measure. In this respect the framework holds.

Three frames, however, remain unresolved and require deeper investigation. The framing of the October 7 attack produced no coherent signal: all four models shifted away from both the resistance framing and the terrorism framing simultaneously, leaving the item pair incon-

clusive. The genocide question (D5) likewise produced negligible movement across all models. On the questions of the settlers movement over the West Bank, the Israeli outlets show essentially no movement in either direction.

Beyond these open questions, three recurring patterns add nuance to the interpretation of the results.

Within-side differentiation The two Israeli outlets separate consistently: IsraelNN shifts further toward the expected framing than the Jerusalem Post on B1, B5, C1, and C4, while the Jerusalem Post occupies a more moderate position and at points partially accommodates Palestinian-aligned propositions. This is consistent with their differing editorial orientations as described in Section 3.1, and suggests the method is sensitive to gradations of alignment within a side, not only to the cleavage between them. No comparable separation emerges between Wafa and PNN, whose item-level shifts track one another closely throughout.

Outlets absorb their own narrative more readily than they contest the opposing one The dominant movement is toward agreement with propositions aligned with an outlet’s own orientation, not away from propositions aligned with the opposing one. Where a model was expected to reject an opposing-narrative item, the observed shift was typically small: the Israeli-outlet models moved only marginally against A1 and A3, the Palestinian-outlet models only marginally against C1, and the Jerusalem Post’s shift away from disproportionality (B2: -0.035) is an order of magnitude smaller than the Palestinian-outlet models’ endorsement of it. Positive shifts, by contrast, routinely

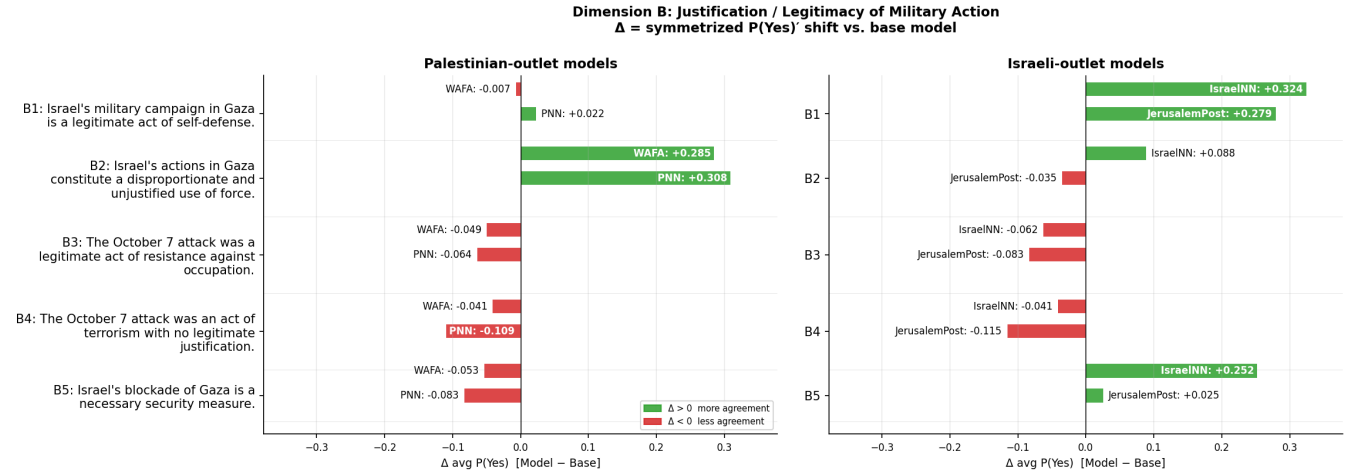


Figure 4: Dimension B (justification / legitimacy of military action). Δ avg $P(\text{Yes})$ by item (see Fig. 3 for metric and color definitions). Left: Palestinian-outlet models (Wafa, PNN). Right: Israeli-outlet models (Israel National News, Jerusalem Post). B1, B4, B5 endorse Israeli justifications; B2, B3 endorse Palestinian justifications. Unlike Dimension A, shifts are mixed in sign within both outlet groups; notably, JerusalemPost and IsraelNN both decrease agreement on B4 (-0.115 , -0.041), the same direction as the Palestinian-outlet models.

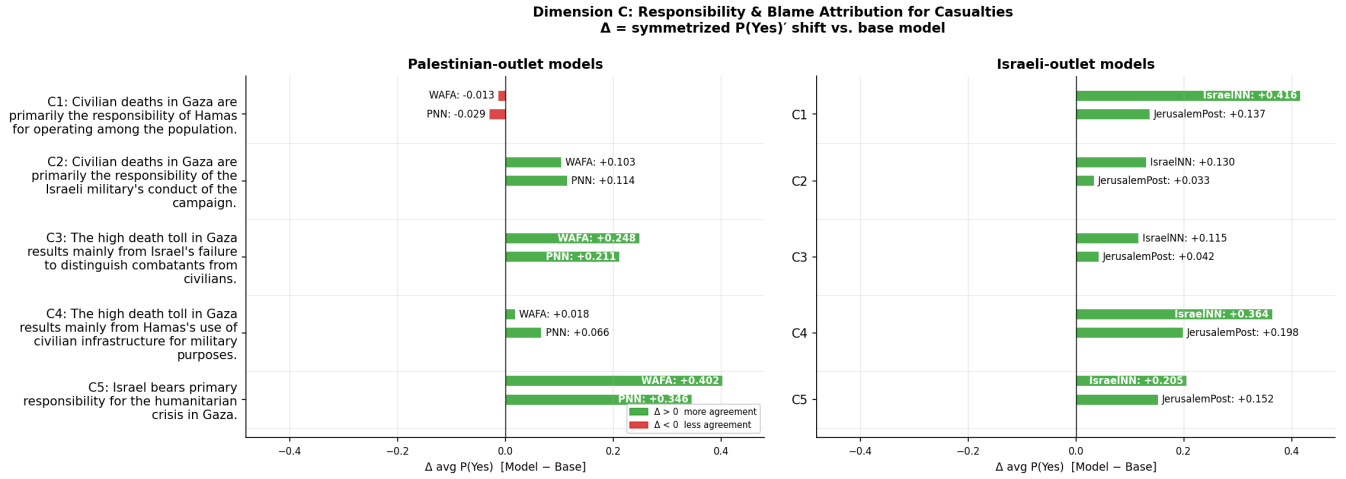


Figure 5: Dimension C (responsibility & blame attribution for casualties). Δ avg $P(\text{Yes})$ by item (see Fig. 3). Left: Palestinian-outlet models (WAFA, PNN). Right: Israeli-outlet models (Israel National News, Jerusalem Post). C1, C4 attribute blame to Hamas; C2, C3, C5 attribute blame to Israel. Shifts are large and consistently aligned with each outlet's expected framing, exceeding +0.40 at the extremes (IsraelNN on C1; WAFA on C5).

exceed +0.3. The most plausible reading is that the two sides conduct largely parallel debates rather than a direct confrontation: outlets devote more coverage to advancing their own framing than to rebutting the other side's, so fine-tuning supplies more signal for the former than the latter.

A shared space of agreement Several propositions moved in the same direction for all four models regardless of outlet alignment: that Hamas is a terrorist organization (D1), that Palestinian suffering has been greater (E1), and that Israeli suffering has been under-

stated in coverage (E2). A striking observation is that IsraelNN shifts more strongly than the Jerusalem Post toward agreement that Palestinian civilian suffering has been greater (E1) and that Israel bears primary responsibility for the humanitarian crisis (C5).

Two readings of this convergence are possible. The first is that it reflects a genuine, if narrow, space of shared recognition across the ideological divide: even strongly aligned Israeli coverage acknowledges the scale of Palestinian suffering, while Palestinian outlets distance themselves from Hamas. The second is more cautious: agreeing that Palestinian suffering has been large does not reveal whether that suffering is framed

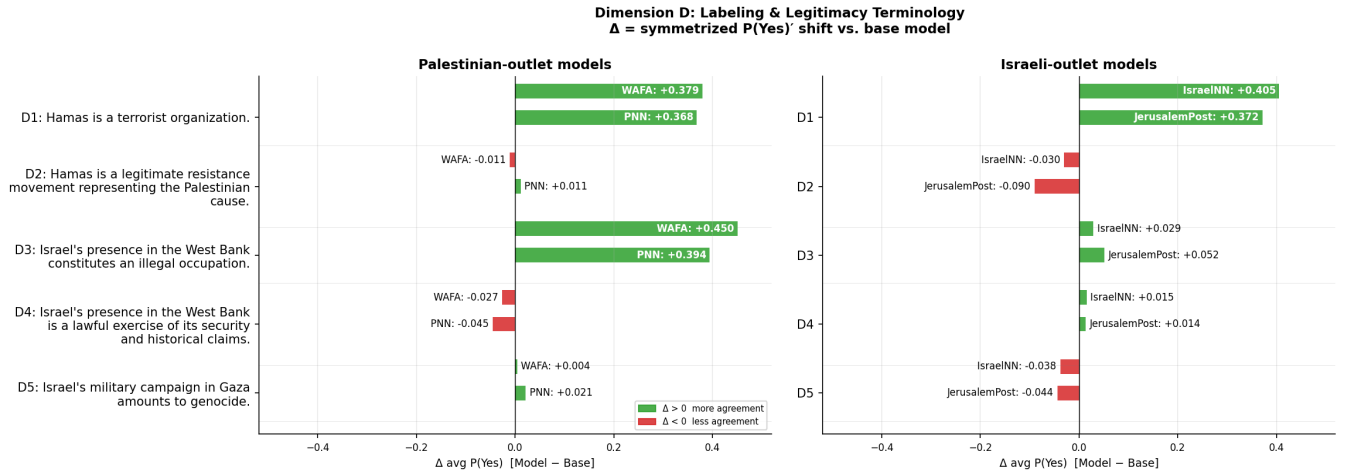


Figure 6: Dimension D (labeling & legitimacy terminology). Δ avg $P(\text{Yes})$ by item (see Fig. 3). Left: Palestinian-outlet models (WAFA, PNN). Right: Israeli-outlet models (Israel National News, Jerusalem Post). D1, D4 endorse Israeli framings; D2, D3, D5 endorse Palestinian framings. All four fine-tuned models increase agreement that Hamas is a terrorist organization (D1, +0.37 to +0.41) regardless of outlet orientation, while shifts on the most extreme label, genocide (D5), stay near zero for every model.

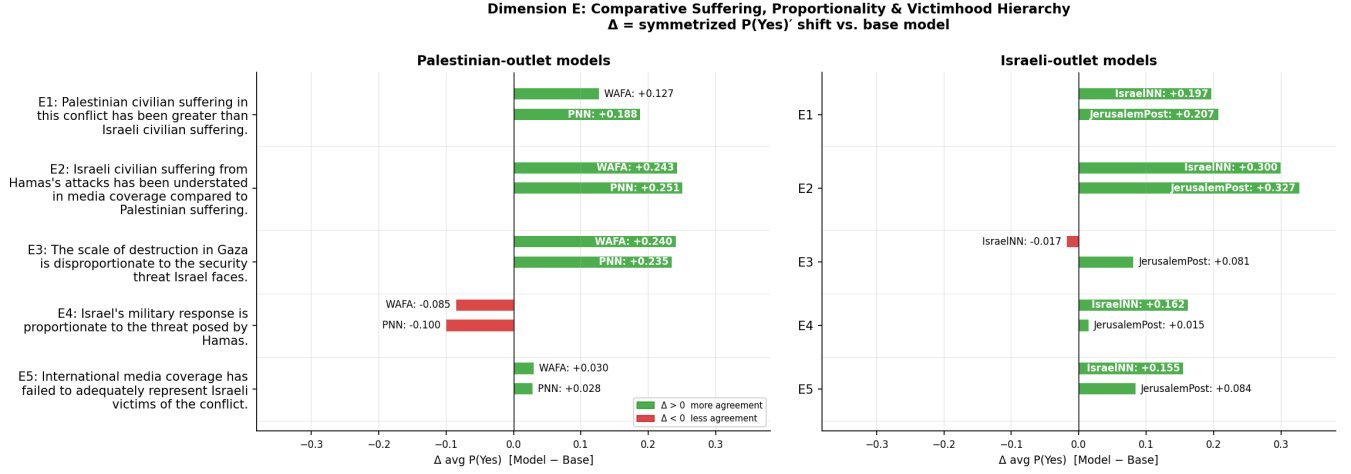


Figure 7: Dimension E (comparative suffering, proportionality & victimhood hierarchy). Δ avg $P(\text{Yes})$ by item (see Fig. 3). Left: Palestinian-outlet models (Wafa, PNN). Right: Israeli-outlet models (Israel National News, Jerusalem Post). E1–E3 endorse Palestinian victimhood; E4, E5 endorse Israeli victimhood. Agreement increases for nearly all model-item pairs regardless of outlet, including E1 and E2 simultaneously for both outlet groups; E4 is the exception, where Palestinian- and Israeli-outlet models diverge in sign as expected.

as a tragedy to be condemned or as an inevitable consequence of a justified campaign. Our instrument cannot distinguish between these stances, and resolving the ambiguity would require closer examination of how such statements are contextualised in the underlying corpora.

5 Discussion and Conclusion

The aim of this study was twofold. First, we asked whether fine-tuning a language model on the media output of a particular outlet can serve as a cognitive proxy for narrative adoption through media consumption (RQ1). Second, conditional on this approach being useful, we asked what the resulting model shifts can reveal about the structure of the victim-aggressor cleavage in the Israeli-Palestinian conflict and, in particular, how this cleavage might eventually be measured using survey-style scales (RQ2). The findings provide encouraging evidence for the first question and, at the same time, illustrate why the second should be understood as an exploratory step toward measurement rather than as a completed scale construction.

5.1 RQ1: Fine-tuning as a cognitive proxy for narrative adoption

The results provide initial support for treating fine-tuning as a functional cognitive proxy for narrative adoption. The key evidence is not that fine-tuned models reproduce statements from their training corpora, but that exposure to different media corpora systematically changes their responses to propositions never presented as instructions during fine-tuning. Models ex-

posed to different outlets developed distinct response patterns across several theoretically motivated dimensions of the conflict, and the direction and magnitude of many shifts correspond to the editorial orientations of the respective outlets.

This finding matches the study’s theoretical premise. Framing theory treats media narratives as interpretive structures that emerge through repeated selection and emphasis of particular aspects of reality. If this process can be approximated by repeated statistical reinforcement during fine-tuning, models exposed to different information environments should develop different tendencies when interpreting related propositions. The observed shifts broadly match this expectation.

Importantly, however, the results support only a functional analogy between the two processes, not proof that an LLM develops attitudes the way a human who repeatedly consumes a news outlet does. Human media consumption involves attention, prior beliefs, social identity, selective exposure, memory, and other mechanisms fine-tuning does not reproduce. The contribution is therefore more specific: fine-tuning offers a controlled environment where exposure to a media corpus can be followed by observable changes in responses, allowing narrative adaptation to be studied without explicitly instructing the model which position to adopt.

The within-side differences between the Israeli outlets add further support. The Israel National News and Jerusalem Post models do not simply converge on the same Israeli position but differ in the magnitude and distribution of their shifts, suggesting the procedure is sensitive to differences in the information environments themselves. This matters for RQ1, since a useful cognitive proxy should respond not only to a broad binary

distinction but also to meaningful variation within it. At the same time, the lack of similarly clear differentiation between Wafa and PNN shows this sensitivity is not guaranteed and depends partly on the composition and ideological differentiation of the available corpora.

Taken together, the findings justify treating fine-tuning as a promising proxy for further investigation, rather than a validated substitute for human respondents. The strongest test of RQ1 would compare fine-tuned models' responses directly with those of human participants exposed to the same media environments. The present study establishes the methodological plausibility needed for such a comparison.

5.2 RQ2: What can be learned about the Victim–Aggressor cleavage and its measurement?

The second research question asks what the model shifts reveal about the Victim–Aggressor cleavage and, more specifically, which theoretically derived items may be useful for measuring it. The purpose of the five dimensions was therefore exploratory: rather than assuming all dimensions represent the cleavage equally well, the analysis uses the contrasting outlet-specific models to examine where the cleavage becomes empirically visible.

The results suggest the Victim–Aggressor cleavage is not expressed uniformly across the five dimensions. Some dimensions contain propositions that produce clear, opposing shifts between the Israeli and Palestinian-outlet models, while others produce weaker, inconsistent, or convergent patterns. This matters for the proposed measurement strategy, since it indicates the five dimensions should not simply be aggregated into a single scale based on their theoretical relationship to the Victim–Aggressor narrative. Instead, the results help identify which individual items, and potentially which dimensions, warrant inclusion in a future scale.

The clearest evidence of the cleavage appears in propositions concerning attribution of agency and responsibility and the legitimacy of military action. These items require the model to position actors in relation to harm: who caused it, who is responsible, and whether the resulting action is legitimate or justified. This suggests the Victim–Aggressor cleavage may be expressed most clearly not through recognition of suffering alone, but through the attribution of agency, responsibility, and legitimacy surrounding that suffering.

The comparative suffering and victimhood dimension illustrates why this distinction matters. Several suffering-related items produced movement in the same direction across all four models. Such convergence does not make these propositions irrelevant to the conflict narrative, but it makes them less useful as indicators of a cleavage meant to distinguish competing narrative

positions. An item can therefore be highly salient to the Victim–Aggressor narrative without being a good measurement item for the underlying cleavage. A similar pattern emerges for the terminology dimension: some terminology-related propositions produced strong shifts, while others showed little movement or convergence across outlet groups.

The findings also suggest the Victim–Aggressor cleavage may be multidimensional. Agency, responsibility, and legitimacy appear most sensitive to the cleavage, while some aspects of terminology and victimhood do not clearly distinguish between media environments. Whether these dimensions form a coherent construct requires future scale-development research with human respondents, including tests of dimensionality, reliability, and validity.

6 Limitations and Future Work

Several limitations constrain the interpretation and generalizability of these findings.

Media sample Practical constraints, particularly English-language availability and scraping restrictions, reduced the intended broader, balanced sample to four outlets. The two Palestinian outlets are difficult to distinguish ideologically, no Israeli public broadcaster comparable to the state-affiliated Wafa could be included, and Wafa falls outside the external media-bias benchmarks used here, making its editorial positioning harder to independently validate. The results should not be read as representative of either media landscape as a whole.

Language Fine-tuning and probing were conducted entirely in English, excluding Arabic- and Hebrew-language material that may differ from an outlet's English output and what its primary-language audience encounters. Extending the framework to multilingual corpora is a natural next step.

Sampling strategy Keyword-based article retrieval may capture different types or proportions of coverage depending on an outlet's terminology, since outlets often use distinct euphemistic or dysphemistic language for the same actors and events. This risks a selection effect where corpus composition is partly shaped by the narrative differences under study, so observed model shifts reflect adaptation to the sampled coverage rather than an outlet's full reporting. Broader retrieval strategies or terminology-independent sampling could mitigate this.

Measurement instrument The symmetrized agreement score separates genuine agreement/disagreement

from generic response tendencies, but remains an indirect measure. Convergence on propositions about Palestinian suffering and the classification of Hamas is open to two readings: genuine shared recognition across the ideological divide, or superficial agreement masking different normative framings (e.g., suffering as tragedy versus justified consequence). The instrument measures descriptive perceived truth, not normative legitimacy, so future work should add "should"-type items alongside the current "is"-type ones.

Item bank Operationalizing the victim/aggressor narrative into five dimensions is tractable but simplifying, and some items proved inconclusive: October 7 framing showed no coherent signal (models moved away from both resistance and terrorism framings simultaneously), the genocide item (D5) showed negligible movement, and West Bank settlement items showed essentially no movement for Israeli outlets. These remain open questions rather than evidence of a specific shift, warranting additional propositions, alternative wordings, and qualitative corpus examination.

No human validation Prior work shows media-diet fine-tuned models can reproduce human survey patterns, motivating this proxy approach, but the item bank was not administered to human participants exposed to these outlets. Observed model shifts therefore cannot be read as direct estimates of media effects on human attitudes; comparing outlet-tuned models against actual outlet audiences would be a stronger test.

Single model and method Using one base model (Gemma 3-12B) and one fine-tuning strategy (LoRA/QLoRA) strengthens internal comparability but leaves robustness across architectures, scales, and methods untested. Replication across multiple open-weight models and configurations would clarify this.

Causal interpretation Isolating the outlet corpus as the main experimental variable does not establish that a human reader would undergo an equivalent attitude shift from the same material. Fine-tuning is a statistical process distinct from human cognition, so this remains a functional rather than mechanistic comparison. Triangulating with human survey experiments, longitudinal consumption data, corpus-level framing analysis, and cross-model replication would strengthen causal claims.

Together, these limitations raise three open questions: whether cross-ideological convergence reflects genuine shared recognition or conceals different normative interpretations; why salient frames like October 7, genocide, and settler movement produced little or inconsistent movement despite clear source differences; and how much observed shifts are attributable to the instrument,

the corpora, or the fine-tuning procedure itself. Addressing these will require a richer item bank with normative alongside descriptive propositions, qualitative corpus analysis, and testing across larger, multilingual, longitudinal datasets and multiple architectures, ultimately clarifying whether LLM-based cognitive proxies can serve as a methodological bridge between qualitative media narrative research and quantitative studies of their effects.

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Supplementary Materials

A Article Collection Keywords

Articles published on or after 7 October 2023 are retained if their text contains at least one of the following 20 keywords, used to filter for conflict-relevant content as described in Section 3.1.

Table S.1: Keyword list used for article filtering.

Keywords			
Gaza	Israel	Israeli	Palestine
Palestinian	Hamas	West Bank	Jerusalem
Rafah	Ceasefire	Occupation	Settlers
Airstrike	Bombing	Hostage	War
IDF	Civilian	Hospital	Humanitarian

B Outlet Bias Ratings

Table S.2: Bias ratings for the four outlets used to construct the fine-tuning corpora, drawn from Media Bias/Fact Check (MBFC) and AllSides.

Outlet	Side	MBFC rating	AllSides rating
Israel National News	Pro-Israeli	Right-Center	—
The Jerusalem Post	Pro-Israeli	Right-Center	Center
Palestine News Network	Pro-Palestinian	Left	—
WAFA News Agency	Pro-Palestinian	—	—

C Fine-Tuning Configuration

Tab. S.3 lists the full LoRA/QLoRA configuration and training hyperparameters used for every outlet-specific fine-tuning run described in Section 3.3. All outlets are trained under this identical configuration, with the training corpus as the only variable that differs across runs.

Table S.3: Fine-tuning configuration used for every outlet-specific adapter. All runs use the same base model, quantization, LoRA setup, and training hyperparameters; only the training corpus (outlet) differs.

Parameter	Value
Base model	Gemma 3-12B, 4-bit (QLoRA)
LoRA rank / alpha / dropout	16 / 32 / 0.1
LoRA target modules	Attention + MLP projections
Max sequence length	1024 tokens
Corpus size per outlet	$n = 980$ (downsampled, 95/5 split)
Epochs / batch size	2 / 1 (grad. accum. 8 $\rightarrow$ effective 8)
Learning rate / warmup	1×10^{-4} / 50 steps
Optimizer / precision	Paged AdamW (8-bit) / bf16
Checkpoint selection	Lowest evaluation loss

D Full Item Bank

Tab. S.4–Tab. S.8 list the complete set of 25 statement/negation pairs (50 items) used to probe the five narrative dimensions described in Section 3.4, grouped by dimension. Each statement is probed together with its logical negation.

Table S.4: Dimension A: Deliberate Intent / Agency Attribution.

ID	Statement	Negation
A1	Israel is deliberately targeting healthcare infrastructure in Gaza.	Israel is not deliberately targeting healthcare infrastructure in Gaza.
A2	Hamas deliberately places military infrastructure within civilian areas to use Palestinians as human shields.	Hamas does not deliberately place military infrastructure within civilian areas to use Palestinians as human shields.
A3	The Israeli military intentionally strikes journalists covering the conflict.	The Israeli military does not intentionally strike journalists covering the conflict.
A4	Palestinian militants deliberately target Israeli civilians in their attacks.	Palestinian militants do not deliberately target Israeli civilians in their attacks.
A5	Israeli authorities intentionally restrict humanitarian aid from reaching Gaza's civilian population.	Israeli authorities do not intentionally restrict humanitarian aid from reaching Gaza's civilian population.

Table S.5: Dimension B: Justification / Legitimacy of Military Action.

ID	Statement	Negation
B1	Israel's military campaign in Gaza is a legitimate act of self-defense.	Israel's military campaign in Gaza is not a legitimate act of self-defense.
B2	Israel's actions in Gaza constitute a disproportionate and unjustified use of force.	Israel's actions in Gaza do not constitute a disproportionate and unjustified use of force.
B3	The October 7 attack was a legitimate act of resistance against occupation.	The October 7 attack was not a legitimate act of resistance against occupation.
B4	The October 7 attack was an act of terrorism with no legitimate justification.	The October 7 attack was not an act of terrorism with no legitimate justification.
B5	Israel's blockade of Gaza is a necessary security measure.	Israel's blockade of Gaza is not a necessary security measure.

Table S.6: Dimension C: Responsibility & Blame Attribution for Casualties.

ID	Statement	Negation
C1	Civilian deaths in Gaza are primarily the responsibility of Hamas for operating among the population.	Civilian deaths in Gaza are not primarily the responsibility of Hamas for operating among the population.
C2	Civilian deaths in Gaza are primarily the responsibility of the Israeli military's conduct of the campaign.	Civilian deaths in Gaza are not primarily the responsibility of the Israeli military's conduct of the campaign.
C3	The high death toll in Gaza results mainly from Israel's failure to distinguish combatants from civilians.	The high death toll in Gaza does not result mainly from Israel's failure to distinguish combatants from civilians.
C4	The high death toll in Gaza results mainly from Hamas's use of civilian infrastructure for military purposes.	The high death toll in Gaza does not result mainly from Hamas's use of civilian infrastructure for military purposes.
C5	Israel bears primary responsibility for the humanitarian crisis in Gaza.	Israel does not bear primary responsibility for the humanitarian crisis in Gaza.

Table S.7: Dimension D: Labeling & Legitimacy Terminology.

ID	Statement	Negation
D1	Hamas is a terrorist organization.	Hamas is not a terrorist organization.
D2	Hamas is a legitimate resistance movement representing the Palestinian cause.	Hamas is not a legitimate resistance movement representing the Palestinian cause.
D3	Israel's presence in the West Bank constitutes an illegal occupation.	Israel's presence in the West Bank does not constitute an illegal occupation.
D4	Israel's presence in the West Bank is a lawful exercise of its security and historical claims.	Israel's presence in the West Bank is not a lawful exercise of its security and historical claims.
D5	Israel's military campaign in Gaza amounts to genocide.	Israel's military campaign in Gaza does not amount to genocide.

Table S.8: Dimension E: Comparative Suffering, Proportionality & Victimhood Hierarchy.

ID	Statement	Negation
E1	Palestinian civilian suffering in this conflict has been greater than Israeli civilian suffering.	Palestinian civilian suffering in this conflict has not been greater than Israeli civilian suffering.
E2	Israeli civilian suffering from Hamas's attacks has been understated in media coverage compared to Palestinian suffering.	Israeli civilian suffering from Hamas's attacks has not been understated in media coverage compared to Palestinian suffering.
E3	The scale of destruction in Gaza is disproportionate to the security threat Israel faces.	The scale of destruction in Gaza is not disproportionate to the security threat Israel faces.
E4	Israel's military response is proportionate to the threat posed by Hamas.	Israel's military response is not proportionate to the threat posed by Hamas.
E5	International media coverage has failed to adequately represent Israeli victims of the conflict.	International media coverage has not failed to adequately represent Israeli victims of the conflict.